An enzyme-discovery agent for de novo biocatalyst search

Target chemistry in → ranked, mechanism-annotated candidates out → structure-ready handoff

THE PROBLEM

Find enzymes for nylon-6,6

• synthesis: adipate + HMDA → amide

• hydrolysis: break the N66 amide

no known natural catalyst — must search

THE FRAMEWORK

target reaction contract

▼

literature + family expansion

▼

decomposed ranking

▼

mechanism + cofactor handoff

▼

structural modeling

WHAT'S NEW

Deterministic mechanism bridge:

activation mode + reactive P + cofactors

derived from the reaction, provenanced

— no LLM, fixes family-keyword errors

KEY RESULTS (run 20260914_220431, assayed panels, exhaustive audit)

1,858

candidates discovered

across 51 families

38 → 20

ranked → structure

handoff, provenanced

52 / 53

mechanism calls agree;

ddaG mis-call fixed

AUROC 0.81

step-1 NAC vs assay

(p = 0.036)

2 targets

synthesis + hydrolysis

same framework

OPEN / HONEST LIMITATIONS

- Ground-truth SfaB is unreviewed (no EC) → outside any reviewed-DB sweep; needs the literature/unannotated channel.
- Improved run spent 72% of tokens idle (a convergence/stop regression to fix).
- Geometry (angle/NAC) is a coarse negative filter, not a quantitative activity predictor.